

MuSR-Stress: A Benchmark for Long-Context and Revisable Belief Reasoning

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March 1, 2026

Abstract

We present **MuSR-Stress**, a benchmark for evaluating reasoning systems along two axes: long-context reasoning and dynamic belief consistency under streaming evidence and counterfactual correction. The benchmark includes two tracks (`long_context` and `dynamic_belief`) with standardized train/dev/test splits. We report metrics beyond final-answer accuracy, including update consistency, calibration (Brier score), required belief revision, recovery rate, and recovery latency. Baseline results show high trajectory consistency but weaker revision success, suggesting that counterfactual corrections remain challenging for current models.

1 Motivation

State-of-the-art language models perform strongly on static QA, but many real settings require reasoning over evolving evidence. Models should maintain coherent beliefs and revise them when new evidence invalidates prior conclusions.

MuSR-Stress targets this gap by jointly evaluating:

- long-context integration under distractor noise,
- dynamic belief revision under streamed and counterfactual evidence.

2 Background & Related Work

MuSR foundation. MuSR introduced structured multi-step soft reasoning with logic-tree-based supervision and narrative contexts. MuSR-Stress extends this setup from static one-shot QA to trajectory-level belief tracking and revision.

Tracks in the current benchmark. The current benchmark provides two tracks:

- `long_context`: stresses long-horizon context integration by injecting distractor cases around the target narrative.
- `dynamic_belief`: unified dynamic track mixing stream-only and counterfactual cases, testing sequential belief updates and correction.

Current scale. Each track has 250 examples split into 175 train / 37 dev / 38 test.

Dynamic-belief format. Each instance contains a question, candidate suspects, gold answer, ordered evidence rounds, and metadata fields such as `has_counterfactual`, `flip_required`, and correction onset round.

3 Proposed Method & Experiments

3.1 Current Pipeline

Seed Sampling (Madlib pools) → Per-Suspect Logic Tree Construction → Validator-Gated Recursive Completion → Murderer/Innocent Chapter Generation → Story Assembly + QA Instance → Stressor Builders (long_context, dynamic_belief) → Train/Dev/Test Split + Evaluation

Figure 1: Current MuSR-Stress pipeline implemented in this repository.

3.2 Generation Settings

The dynamic track is generated with configurable round limits and correction rates. Current v1 settings:

- `max_rounds=40`
- `setup_sentences=2`
- `counterfactual_rate=0.7`
- `flip_rate=0.7`

This produces both:

- **stream-only** cases (no contradiction correction), and
- **counterfactual** cases (with a correction block that may require answer revision).

3.3 Example: Counterfactual Case

Table 1 shows a representative `flip_required` counterfactual instance from the test split. Suspects are Milton and Emma; the victim is Stella, murdered with a nunchaku in her kitchen. Early rounds build evidence against Emma (means via martial arts training, opportunity via scheduled kitchen fitting). The counterfactual block at rounds 21–22 withdraws the key witness timeline against Emma and introduces new physical evidence implicating Milton. Post-correction evidence continues, requiring the model to maintain the revised belief through additional rounds.

3.4 Evaluation Protocol

Let $T = (r_1, r_2, \dots, r_n)$ be the sequence of rounds for a case, $\hat{s}_t$ the model’s top suspect at round t , $p_t(s)$ the model’s probability for suspect s at round t , s^* the gold suspect, and r_{cf} the counterfactual onset round (where applicable). For counterfactual cases, the *active gold* switches at r_{cf} : $g_t = s_{\text{before}}^*$ if $t < r_{cf}$, else $g_t = s_{\text{after}}^*$.

Final accuracy.

$$\text{final_accuracy} = \mathbf{1}[\hat{s}_n = s^*]$$

Brier score.

$$\text{BS}_t = \sum_{s \in \mathcal{S}} (p_t(s) - \mathbf{1}[s = g_t])^2$$

$$\text{brier_final} = \text{BS}_n; \text{brier_mean} = \frac{1}{n} \sum_{t=1}^n \text{BS}_t.$$

Table 1: Example counterfactual case (`dynamic_belief_13.q0`). Gold switches from Emma $\rightarrow$ Milton at round 21.

Round	Type	Evidence
1–2	setup	In Mapleton, Stella is murdered by a nunchaku in her kitchen. Emma, a fashion designer, was training in martial arts that morning.
3	tree_fact	Nunchaku are commonly used in martial arts.
5	tree_fact	Milton is a martial arts enthusiast.
8	tree_fact	Emma’s martial arts class included training with nunchaku.
14	tree_fact	Emma was working at the house that day.
18	tree_fact	Emma and Stella were scheduled to meet in the kitchen for a fitting.
21	cf_update	Correction: the key witness timeline against Emma had an incorrect timestamp and is withdrawn.
22	cf_update	Follow-up records place Milton’s phone near the scene during the murder window, show a matching weapon purchase, and include threatening messages to Stella.
23	tree_fact	A local journalist photographed Milton at various crime-linked locations.
27	tree_fact	Milton and Tommy were captured in a photo together at a bar.

Update consistency. Let $\mathcal{F}$ be the set of round indices where $\hat{s}_t \neq \hat{s}_{t-1}$. In flip-required counterfactual cases, the first transition in $\mathcal{F}$ with $t \geq r_{cf}$ and $\hat{s}_t = s_{\text{after}}^*$ is exempted (it is a correct revision). The flip penalty is:

$$\text{flip_penalty} = \frac{|\mathcal{F}_{\text{penalized}}|}{n - 1}$$

For the trajectory penalty, let τ be the subsequence of rounds used (all rounds for stream-only; post- r_{cf} rounds for counterfactual cases):

$$\text{traj_penalty} = \frac{\sum_{t \in \tau, t > 1} \max(0, p_{t-1}(\hat{s}_n) - p_t(\hat{s}_n))}{|\tau| - 1}$$

$$\text{update_consistency} = \max(0, \min(1, 1 - \frac{1}{2}(\text{flip_penalty} + \text{traj_penalty})))$$

Flip when required. Defined only for flip-required counterfactual cases. Let pre- and post-correction predictions be $\hat{s}_{\text{pre}} = \hat{s}_{r_{cf}-1}$ and $\hat{s}_{\text{post}} = \hat{s}_n$:

$$\text{flip_when_required} = \mathbf{1}[\hat{s}_{\text{post}} = s_{\text{after}}^*]$$

Stability when not required. Defined only for counterfactual cases where `flip_required=False`:

$$\text{stability_when_not_required} = \mathbf{1}[\hat{s}_{\text{pre}} = \hat{s}_{\text{post}}]$$

Recovery rate and latency. Defined only for flip-required counterfactual cases. Let $t^* = \min\{t \geq r_{cf} : \hat{s}_t = s_{\text{after}}^*\}$ (undefined if never reached):

$$\text{recovery_rate} = \mathbf{1}[t^* \text{ exists}] \quad \text{recovery_latency} = t^* - r_{cf}$$

Recovery latency is reported only over recovered cases (where t^* exists), so a value of 0 indicates immediate correction at the counterfactual round.

Table 2: Dynamic-belief test results ($n = 38$).

Metric	Qwen2.5-7B-Instruct	Llama-3.3-70B-Instruct
Final accuracy	0.789	TBD
counterfactual subset (n=28)	0.893	TBD
stream-only subset (n=10)	0.500	TBD
Update consistency	0.940	TBD
Brier final	0.344	TBD
Brier mean	0.431	TBD
Flip when required	0.952	TBD
Stability when not required	1.000	TBD
Recovery rate	1.000	TBD
Recovery latency (rounds)	0.43	TBD

3.5 Preliminary Baseline Results

We evaluate instruct models at temperature 0 on the `dynamic_belief` test split ($n = 38$).

Key observations. Qwen2.5-7B-Instruct handles counterfactual revision remarkably well: `flip_when_required`=0.952, `recovery_rate`=1.0, and `recovery_latency`=0.43 rounds. This suggests the model responds quickly and reliably to explicit correction signals. The more surprising finding is the gap between counterfactual accuracy (0.893) and stream-only accuracy (0.500): cases *without* a correction are harder than cases *with* one. A likely explanation is that counterfactual cases provide an anchoring signal — the correction text explicitly names the revised suspect — whereas stream-only cases require reasoning purely from accumulated evidence. Note that the stream-only subset is small ($n = 10$), so this gap should be interpreted cautiously.

3.6 Dataset Composition

The full `dynamic_belief` track (250 cases, 70/15/15 train/dev/test split, seed=7) breaks down as follows from the generation manifest:

Table 3: MuSR-Stress `dynamic_belief` track composition.

Split	Total	CF / Stream-only
Train	175	—
Dev	37	—
Test	38	—
Total	250	176 CF / 74 stream-only
flip-required CF	128	(of 176 CF)
flip-not-required CF	48	(of 176 CF)

4 Next Steps

Current limitations include single-domain emphasis and single-seed headline reporting. Next steps are:

- multi-seed reporting with mean/variance,
- failure taxonomy (spurious flip vs missed required flip),
- evaluation of additional models on both tracks (`long_context`, `dynamic_belief`),
- a combined track stressing both long-context attention and dynamic belief revision simultaneously,
- extension of dynamic-belief generation to other domains beyond murder mystery.

5 Conclusion

MuSR-Stress provides a practical benchmark for evaluating not only correctness, but also whether models maintain and revise beliefs appropriately as evidence evolves across two complementary tracks. Results motivate targeted progress on non-monotonic belief updating and long-context attention under distractor noise.